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Unit – 5 : Session – 8 : SLO – 2 | Implementation of Shortest Path Algorithm — Activities

■ I. Algorithm Storyboarding (graph: 0-1(2), 0-2(6), 1-3(5), 2-3(8))

- Panel 1: Selected vertex 0. $\text{dist} = [0, 2, 6, \text{inf}]$. Visited = {0}. Unvisited = {1, 2, 3}.
- Panel 2: Selected vertex 1 (smallest dist). $\text{dist} = [0, 2, 6, 7]$. Visited = {0, 1}. Unvisited = {2, 3}.
- Panel 3: Selected vertex 2. $\text{dist} = [0, 2, 6, 7]$ ($0-2-3 = 14 > 7$, no change). Visited = {0,1,2}. Unvisited = {3}.
- Panel 4: Selected vertex 3. $\text{dist} = [0, 2, 6, 7]$. Visited = {0,1,2,3}. Unvisited = {}. DONE.
- Final shortest distances from 0: $0 \rightarrow 0$, $1 \rightarrow 2$, $2 \rightarrow 6$, $3 \rightarrow 7$ (via $0-1-3$).

■ II. Spot-the-Silent Bug

- Bug: every $\text{dist}[i]$ is initialised to 0, so all vertices look 'already shortest' and no relaxation ever improves them; Dijkstra's returns 0 for every vertex.
- Fix: initialise $\text{dist}[i] = \text{INT_MAX}$ (infinity) for all $i \neq \text{src}$; only $\text{dist}[\text{src}] = 0$.

■ III. Challenge-The-Logic — Claim: 'with all edge weights = 1, Dijkstra's behaves like BFS'

- TRUE. With unit weights, the priority queue always pops vertices in order of increasing hop-count, which is exactly the order a queue produces in BFS.
- Example: edges $0-1(1)$, $0-2(1)$, $1-3(1)$. BFS from 0 visits 0,1,2,3. Dijkstra's distances: 0,1,1,2 —
- vertices are finalized in the same order 0,1,2,3.

■ IV. Real-World Map Simulation (Campus)

- Places: Main Gate (G), Library (L), Canteen (C), Class Block (B).
- Edges (minutes): G-L 5, G-C 2, C-L 1, C-B 4, L-B 3.
- Start: Main Gate. Dijkstra's:
- dist : $G=0$, $C=2$ (via G-C), $L=3$ (via G-C-L, better than $G-L=5$), $B=6$ (via G-C-L-B).
- Final distance table: $G \rightarrow G = 0$, $G \rightarrow C = 2$, $G \rightarrow L = 3$, $G \rightarrow B = 6$.

■ V. Run It Backwards

- Given $\text{dist}[0]=0$, $\text{dist}[1]=3$, $\text{dist}[2]=5$, one possible edge set:
- $0-1$ weight 3, $1-2$ weight 2 (so $0 \rightarrow 1 = 3$, $0 \rightarrow 2 = 3+2 = 5$).
- Alternative: $0-2$ weight 5 and $0-1$ weight 3 with no cheaper $0 \rightarrow 2$ route.